

Dú 2.1

Nechť $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ je matice definující $\langle, \rangle$

$$B = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right)$$

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = (1, 0) \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = a$$

$$\left[\begin{array}{l} \text{2. Tvrzení 8.50 : } \langle u, w \rangle = [u]_B^* [w]_B \\ \text{2. Tvrzení 8.48 : } [u]_B = (\langle v_1, u \rangle, \langle v_2, u \rangle, \dots) \end{array} \right]$$

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = \left(\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle, \left\langle \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle \right) \cdot \left(\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle, \left\langle \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle \right)^T$$

$$a = (1, 3) \begin{pmatrix} 1 \\ 3 \end{pmatrix} = 10$$

Identickým postupem* zjistíme i b, c, d a získáme.

$$A = \begin{pmatrix} 10 & 23 \\ 23 & 53 \end{pmatrix}$$

*

$$\left(\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle, \left\langle \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle \right) = (2, 7)$$

$$b = \begin{pmatrix} 1 \\ 3 \end{pmatrix} (2, 7) = 23$$

$$c = \begin{pmatrix} 2 \\ 7 \end{pmatrix} (1, 3) = 23$$

$$d = \begin{pmatrix} 2 \\ 7 \end{pmatrix} (2, 7) = 53$$